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GUIDELINES TO CLINICAL TRIALS IN PREGNANCY AND LACTATION

This document has been prepared to serve as a guideline for the preparation of applications to conduct clinical trials in South Africa. It represents the South African Health Products Regulatory Authority's [SAHPRA] current thinking on pregnancy and lactation for clinical research in South Africa. It is not intended as an exclusive approach, and SAHPRA reserves the right to request any additional information and may make amendments in keeping with the knowledge that is current at the time.

Document History

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Glossary

Abbreviation/ Term	Meaning
ABPI	The Association of the British Pharmaceutical Industry
CIOMS	Council for international organizations of medical sciences
COVID	The disease Covid-19
CTC	Clinical Trial Committee
CTF1	Clinical trials application form
DSMB	Data and Safety Monitoring Board
HIV	Human immunodeficiency virus
IP/IMP	Investigational product/ Investigational Medicinal Product
IVDs	In vitro diagnostic medical devices
LMIC	Low- and Middle-Income Countries
NCE	New chemical entity
UN	United Nations
PK and PD	Pharmacokinetics and Pharmacodynamics
SAD & MAD	Single Ascending Dose and Multiple Ascending Dose
SAGCP	The South African Good Clinical Practice guidelines
SAE/AESI/AEs	Serious adverse event/ adverse event of special interest/ adverse events
SAHPRA	The South African Health Products Regulatory Authority
SOP	Standard Operating Procedure
TIE	Time, Inconvenience and Expenses

1. INTRODUCTION

All clinical trials related to pregnancy and lactation need to be rigorously scientifically, ethically, statistically, and clinically sound and appropriate to both the needs of this group of participants and important to the needs of the South African and the rest of the world thereby ensuring maximal safety and minimal risk as well as critical benefits to Women of childbearing age, fetuses and neonates/ infants.

These studies therefore need to be in conformance with the latest and current guidelines regarding Good Clinical Practice SAGCP, National Department of Health Guidelines for Ethics and Research as well as the latest Declaration of Helsinki, ICH-GCP and CIOMS as well as other relevant clinical research guidelines.

Clinical research with individuals, groups, or communities in situations of vulnerability is only justified if it is responsive to both their health needs and priorities as well as those of groups and communities who stand to benefit from the resulting knowledge, practices, or interventions.

Such studies further need to conform to the UN Universal Declaration of Human Rights, the Constitution of the Republic of South Africa, and relevant laws and regulations in the Republic.

While pregnancy is not considered a disease or illness, it needs to be recognized that there are physiological and psychological changes in pregnancy, foetal and neonatal/ infant milieus, which may exhibit unique PK and PD parameters. There may be unique pathological conditions contributing to the need to study interventions in these participants.

1.1 Purpose

There is a need to be more inclusive, and retention of, with pregnancy and lactation related interventional trials involving investigational products (new and existing chemical entities) and medical devices and IVDs whilst not compromising safety, participant benefit, and protection of the participant, in all studies, Phases 1 to 4. Studies in this domain can be planned either as independent study or a sub study or as continued pregnancy follow up embedded within studies or sub studies with other objectives.

1.2 Scope

This guideline applies to all clinical trials (Phase 1-4). It establishes a framework for:

- Protection of participants in clinical trials related to pregnancy and lactation (maternal, foetal and neonatal/ infant).
- Enabling appropriate conduct and successful outcome of clinical trials in pregnancy and lactation.

- To guide stakeholders on clinical trials during pregnancy and lactation.
- To guide management of participants' outcomes for participants who become pregnant and/or lactate during clinical trials not primarily designed for reproductive health.

In case of studies involving medical devices/ IVDs (e.g., diagnostic assays and kits, monitoring systems), refer also to the Guidelines on clinical investigation requirements on <https://www.sahpra.org.za/medical-devices-and-in-vitro-diagnostics-guidelines>.

2. LEGAL PROVISION

These guidelines are established in terms of Regulations issued in sections 90(s) and 72(6)(c) of the National Health Act, Act No. 61 of 2003, as well as Regulation 30 of the Medicines and Related Substances Act, Act No. 101 of 1965.

All role players involved in clinical trials related to pregnancy and lactation should also be familiar with the Constitution of the Republic of South Africa, national and international guidelines, including but not limited to the following current versions or their successors:

- SA GCP
- South African Ethics in Health Research Guidelines: Principles, Processes and Structures
- Universal Declaration of Bioethics and Human Rights
- Applicable legislation and associated regulations
- ICH GCP
- Declaration of Helsinki
- ABPI (Section Special Populations: Women)

3. PREGNANCY

3.1 Pre-submission Meetings

Stakeholders should, at both regulatory request and/ or self-decision, engage with SAHPRA to ensure aspects of the planned study are discussed for smooth progress and regulatory success (refer to communications with SAHPRA guidelines).

Pregnant or breastfeeding women should be included in human clinical trials where possible, i.e., when safety and other relevant information are available during a clinical trial. This is to ensure that appropriate knowledge about NCEs for this group of people is developed.

3.2 Public Health Emergencies

The recent COVID and HIV pandemics have illustrated the need for adequate study and planning of the use of IPs in pregnancy and lactation.

Studies that address the scenarios of public health emergencies will be prioritized at the CTC level to expedite early implementation.

More frequent vigilance in terms of early SAE/ AESI reporting or progress reports or applicant engagement, or interim analyses by DSMBs or sponsors may be required at regulatory discretion.

3.3 Prevailing Evidence (Non-clinical/ Clinical)

In preparation for human studies, non-clinical developmental and reproductive safety and toxicology studies must be done along with genotoxic studies. These studies must be available for any SAHPRA submission for evaluation.

All clinical studies in pregnant women require clinical data in non-pregnant participants before extending to pregnant participants. Outcomes of these studies will guide extension to pregnancy and lactation.

As there are maternal and fetal differences in each trimester, any study needs to take cognisance of the appropriateness of the study design per trimester.

3.4 Clinical studies incorporating pregnancies

In specific, predetermined, pre-discussed situations, clinical studies may progress directly to pregnant and lactating participants.

If the applicant/ sponsor determines that proceeding with trials in pregnancy is appropriate, then the following approaches/ actions (in no specific order) need to be considered and/ or incorporated into the development strategy:

- Recruitment of pregnant individuals in each trimester into ongoing and/ or subsequent clinical trials,
- Removal of mandatory contraception requirements in ongoing and/or subsequent clinical trials,

- Implementation of study(ies) specifically designed to be conducted in pregnant individuals if needed,
- Ongoing participation of individuals who become pregnant during clinical trials (source: ICH E21 Guidelines), and
- Management of pregnancies where contraceptive methods have failed requires a safety review before concluding on the risk-benefit of the IP.

Pregnant individuals must continue in the study but not receive any further IP (unless otherwise specified in the protocol or if there is a demonstrated positive risk-benefit).

The baby should be followed up for an appropriate period, preferably up to a minimum of 18 months.

3.4.1 Scenario of Failed Contraception in Studies where WOCBP are enrolled with mandatory contraception

Women of childbearing potential may experience a “breakthrough pregnancy” despite the implementation of a trial-required method of contraception. Several outcomes may be presented depending on an informed choice and intensive information sharing by the investigator and/ or sponsor, such as:

- Continuation of pregnancy with or without IP under strict(er) investigator/ sponsor guidance and surveillance until birth and close evaluation of infants thereafter,
- Termination of Pregnancy and no continuation in the study,
- Possible unblinding,
- Reporting to the regulatory authority,
- Re-evaluation of participant insurance if needed, and
- Pregnant participants should be retained in studies as far as possible, and allowance should be made to incorporate obstetric and post-natal and paediatric care within the study.

3.5 Phase 1 Studies - (Refer to first-in-human, Phase I guideline)

3.5.1 In exploratory Phase 1 studies, (SAD and MAD) women should typically be of “non-childbearing potential”. (ABPI).

3.5.2 Generally, Phase I trials should not be done in pregnant women.

- 3.5.3 Should the trials be required in pregnant women, they must be performed at a later stage of clinical IP(s) development once there is data from non-pregnant participants.

3.6 Procedures

Procedures should be carried out at minimal risk and at the very least within the standards of antenatal care. The following items should be summarized and further described and included in the site capacity as per Section 2.4 of the new trial application (CTF1). The SOP should further be provided for the items below:

- An antenatal and postnatal care plan must be provided,
- Health care professionals experienced in ultrasound should be available,
- Midwives and obstetricians as well as other Health care professionals with adequate experience should be available for delivery,
- Site should include investigators with experience in neonatal management,
- Participants should have easy access to antenatal and labour care,
- Procedural obstetrics (Chorionic villous sampling, amniocentesis) should be done if clinically indicated,
- Cord blood to be taken only after delayed cord clamping,
- Adhere to standardized blood volume draws as per the Ethics committees or SAHPRA CTC review,
- Infant follow-up should be with qualified personnel and
- Appropriate emergency trolley as per SAHPRA guidelines must be available on site

3.7 Sites and Investigators

3.7.1 Clinical trial sites should have investigators who are obstetricians/ diplomates or proficient in the management of obstetric emergencies. Sites should be set up to manage these or have quick and established access to a facility with expertise and capacity to manage such outcomes.

3.7.2 Sonography has become an integral part of the standard of antenatal care- hence demonstrated experience and/or qualification needs to be provided in the CTF1 Section 2.4.

3.7.3 The inclusion of neonatologists and/or paediatricians should be considered where appropriate in the continuum of care.

3.7.4 Any study that involves pregnant women needs to have a formal SOP on where and how deliveries will be managed. The applicant should propose and justify, based on the complexities of the study, whether incorporation of an obstetrician and /or paediatrician and neonatologist may be necessary.

3.8 Registration of Investigational Product

SAHPRA CTC will only consider pregnancy/lactation-related studies where there is a potential benefit to the South African population.

3.9 Foetal/ Neonatal and Infant extension

3.9.1 Many studies in pregnancy and lactation may have objectives and endpoints (or estimands) related to the foetus and neonate or infant.

3.9.2 In studies where there are extensions to monitoring the outcomes of neonates and infants, as incorporated in the study design, these studies should include investigators with experience in paediatrics and/ neonatology and incorporate (both parent/ caregiver) TIE and insurance for the neonate/ infant, as well as clinical care for the duration of the study.

3.10 Pregnancy studies with benefit to the Neonate/ Infant

3.10.1 Pregnancy (or maternal) studies may be justified to have value for maternal outcomes, or maternal-foetal, or maternal-infant outcomes. The Ethics thereof are evolving but advanced, especially as regards vaccine studies in the maternal population. SAHPRA recognizes that IPs are also being evaluated in this population.

3.10.2 SAHPRA CTC encourages the implementation of such studies on CTC review.

3.11 Retention of obstetric and paediatric records

In South Africa, child health records should be retained at least until the child's 21st birthday, as legally, minors have until three years after reaching the age of majority to make claims. This applies to various health records, including those related to obstetric care. Additionally, records should generally be kept for at least six years after they become dormant (date of last contact). Records for patients with mental incapacities should be kept for their lifetime.

3.12 Databases for Long term follow up of babies in the trial

Databases for long term follow up of babies born to pregnant women on clinical studies should be maintained by the sponsor for a period relevant to the study.

4. LACTATION

4.1 Breastfeeding Support

4.1.1 It is recognised that breastfeeding support and successful initiation and prolongation of breastfeeding in South Africa require a sustained collaborative effort to improve, as evidenced by the Tshwane declaration in support of breastfeeding (2011).

4.1.2 In the context of clinical studies, investigational products can potentially be present in breastmilk and affect the neonate or infant. It is thus necessary to include studies or initiate studies on drug metabolism and excretion during lactation for infant safety.

4.1.3 Although it is accepted that “breast is best”, outcomes of lactation research may indicate that, in certain circumstances, alternatives to breastfeeding might be an option to avoid exposure to drugs in breastmilk.

4.1.4 Infrastructural support is strongly recommended by providing safe, private spaces at clinical trial centers with lactation consultant support if needed. Cultural/ Religious and Traditional practices and beliefs must be considered in the design of lactation studies.

4.1.5 If required, the inclusion of paediatricians and lactation consultants may be necessary.

4.1.6 Lactation studies in South Africa and other LMICs should be supported

4.2 Objectives

Ultimately, clinical guidance on drug concentrations in breastmilk is driven by 3 outcomes:

- The concentration of the IP in breastmilk
- The dose of exposure to the infant
- The safety (and possibly benefit) to the infant

Breastmilk is a dynamic human tissue. Characteristics differ from colostrum to foremilk, hindmilk, premature, and term milk, with intraday variations. Thus, it is recommended that protocol norms be established for uniformity in the collection of specimens.

4.3 Inclusion of lactating mothers in studies

- Non-nursing mothers who are still producing breastmilk (but not breastfeeding their infant) may be included in lactation studies.
- Mothers who are breastfeeding: To assess drug exposure in breastmilk and breastfed infants.
- Extension of pregnancy studies to the lactation period.
- Temporary cessation of breastfeeding and replacement with formula will not be allowed.

4.4 Safety and Efficacy monitoring

4.4.1 The same principles apply with respect to the safety monitoring of lactating mothers and their infants. The table of visit schedules should adequately consider this in the protocol.

4.4.2 In addition, certain conditions unique to the neonatal period may require extra consideration (growth, development, neonatal pathology).

4.4.3 Neurodevelopmental outcomes may be of consideration for long-term follow-up.

4.4.4 For studies where there is maternal intervention for infant benefit, such as vaccine studies, more information may be obtained by analysis of breastmilk.

4.4.5 Causality with SAEs/ AEs/ AESIs must be interpreted within the context of plausible exposure to IP.

4.4.6 Termination of studies in lactation will be guided by Safety Review committees, DSMBs, and regulatory authorities.

4.5 Breastmilk Volumes

4.5.1 Standardized reference values for volumes of breastmilk collection in clinical studies are not available.

4.5.2 The minimum amount of breastmilk for the indicated study should be utilized for optimal PK/ PD analysis, or any other analysis.

4.5.3 Breastmilk is considered Human Tissue in South Africa (National Health Act 61 of 2003), and all biological hazard precautions in acquisition, processing, storage, and transport should be taken. An export licence and material transfer agreement will be necessary for the export of the specimen.

NOTE: Only a maximum sample storage of 15 years is allowed unless it is scientifically justified.

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6. VALIDITY

This guideline is valid for a period of 5 years from the effective date of implementation. It will be reviewed on this timeframe or as and when required.